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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S5 (R, S) / S5 (WP) (R) / S3 (PT) (S,FE) Examination November 2024 (2019 Scheme)

Course Code: EET 303

Course Name: MICROPROCESSORS AND MICROCONTROLLERS

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks) Marks

1	How does the 8085 microprocessor handle 16-bit data using its 8-bit registers?	3
2	Describe the operation of the stack in the 8085 microprocessor.	3
3	Classify the 8085 instructions based on the number of bytes and give an example for each type.	3
4	Explain the branching instruction in 8085 with an example.	3
5	What is vectored and non-vectored interrupts in 8085? List it.	3
6	Describe the functions of the following pins in 8051 microcontrollers a) PSEN, b) ALE, c) EA	3
7	Explain the difference between ORG and EQU directives in 8051 programming.	3
8	Identify the addressing mode used in the instruction MOV @R1, A and explain its function.	3
9	Describe the 16-bit timer mode of the 8051's Timer.	3
10	What is the role of the SCON register in the 8051 microcontroller's serial communication? List its bits.	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	Explain the architecture of the 8085 microprocessor with the help of a block diagram.	10
	b)	Describe the operation of the following 8085 instructions: (i) RAR, (ii) DAD, (iii) XCHG, (iv) DAA	4

12	a)	Draw the timing diagram for the LDA 2000H instruction and explain each machine cycle involved in the execution of this instruction.	10
	b)	Identify the machine cycles of the following instructions: (i) STA 3000H, (ii) CPI 10H, (iii) LDAX D, (iv) RRC	4

Module -2

13	a)	Write an 8085-assembly language program to sort the numbers in an array in ascending order	8
	b)	Describe the various types of arithmetic instructions in the 8085 microprocessor. Provide examples.	6
14	a)	Write an ALP of 8085 to create a delay of 10 milliseconds with a clock speed of 3 MHz	7
	b)	Write an 8085 ALP to check if two 8-bit numbers stored at memory locations 2600H and 2601H are equal. Store 01H at 2602H if they are equal, otherwise store 00H.	7

Module -3

15	a)	Interface 4 seven-segment LED with 8085 using 8255, write an ALP to display a 4-digit BCD number.	10
	b)	Describe the RIM instruction in the 8085 microprocessor, explaining its functionality.	4
16	a)	Describe the internal RAM organisation of 8051	8
	b)	Explain the differences between hard and soft real-time systems in embedded applications. Provide two examples of applications for each type	6

Module -4

17	a)	Describe branch control instructions of the 8051 microcontroller in detail.	8
	b)	Describe the addressing modes in the 8051 microcontroller?	6
18	a)	Write an ALP of 8051 to read a 4-bit input from Port 1 (P1.0 to P1.3) and output it to Port 2's upper nibble (P2.4 to P2.7).	6
	b)	Write an Embedded C program to read the state of a push button connected to Port 3, pin 2 (P3.2) and turn on an LED connected to Port 1, pin 0 (P1.0) when the button is pressed.	8

Module -5

19	a)	Explain the different modes of operation of timers in the 8051 microcontroller in detail.	8

	b)	Write a C program for the 8051 microcontrollers to use Timer 1 in 8-bit auto-reload mode to generate a periodic delay of 50 microseconds, XTAL = 12MHz.	6
20	a)	Write an Embedded C program to configure the 8051 for serial communication at a baud rate of 4800. Use Timer 1 for baud rate generation.	6
	b)	Interface DAC 0808 to Port 1 of the 8051 microcontroller. Write an Embedded C program to send a digital value to DAC, to generate an analog output.	8

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